Daniel Lane

Daytona Beach, FL | (386) 748-2720 | DPLane2720@gmail.com
www.dplane.com | linkedin.com/in/dplane2720 | github.com/DPLane2720

Summary

PhD mechanical engineer specializing in real-time sensor fusion and perception, grounded in machine learning, data-driven system identification, and nonlinear dynamics. Designed and field-deployed complete perception stacks on uncrewed vehicles. Background spans enterprise systems leadership, graduate robotics research, and university teaching. Seeking R&D and applied-engineering roles in robotics, perception, and autonomy.

Education

Embry-Riddle Aeronautical University – PhD in Mechanical Engineering — Dec 2025
Dissertation: "A Study in Object Detection and Classification Performance by Sensing Modality for Autonomous Surface Vessels"
Embry-Riddle Aeronautical University – MS in Mechanical Engineering — May 2021
Stetson University – BS in Physics — May 2009

Technical Skills

Languages & Tooling: Python, MATLAB/Simulink, C++, ROS/ROS2, Git, Linux
Perception & Sensor Fusion: LiDAR–camera fusion, object detection & classification (YOLO), multi-sensor calibration (intrinsic/extrinsic), time synchronization (PTP), OpenCV, GStreamer
Machine Learning & Modeling: Neural networks (CNNs), system identification (SINDy), dynamic mode decomposition / Koopman, information-theoretic methods, Kalman filtering, nonlinear dynamics
Applied AI: LLM-agent systems, retrieval/context pipelines, API integration
Robotics & Embedded: Real-time embedded deployment (NVIDIA Jetson AGX Xavier), motion planning (A*/RRT*), Gazebo/RViz, mechatronics, CAD (Fusion 360, SolidWorks)

Engineering Projects & Research

Real-Time LiDAR–Camera Sensor Fusion for Autonomous Perception
ASV Perception Research, Robotics & Autonomous Systems Lab (RASL), ERAU | NEEC/ONR Grant — 2024 – 2025

- Developed a lightweight late-fusion method combining vision (YOLOv8) and LiDAR (GB-CACHE) object detections through Kalman-weighted updates, generalizable across sensor modalities and detection frameworks
- Performed full spatiotemporal calibration across a 7-sensor array, pairing intrinsic/extrinsic spatial alignment with sub-millisecond PTP timestamp synchronization, enabling unified point cloud/image fusion at 5 fps
- Built and operated the field data-collection pipeline aboard the autonomous vessel, capturing 52M+ sensor messages and 2+ TB of synchronized LiDAR/video across 122 ROS bag files over 13 field days

Lead Perception Engineer – RobotX 2024 & RoboBoat 2025
Sr. Software Team Member, Team Minion, ERAU — 2024 – 2025

- Led perception system architecture for an international autonomous-vehicle competition; team placed 3rd overall and 1st in technical documentation at the 2024 Maritime RobotX Challenge
- Developed real-time LiDAR–HDR fusion pipelines for object detection, classification, and color identification supporting autonomous mission planning
- Designed ground control interfaces and sensor pipelines for real-time embedded deployment

Autonomous Vessel Concept Development – Yamaha Marine (NDA)
Co-Researcher, ERAU — 2024

- Collaborated under NDA with Yamaha Marine Division to develop high-level design concepts for future vessels

Information-Theoretic & Data-Driven Model Discovery for System Identification
Researcher, Complex Dynamical Systems Lab (CDSL), ERAU | NSF Grant — 2022 – 2023

- Recovered nonlinear system dynamics directly from data using information-theoretic structure detection and sparse identification (SINDy), resolving interaction directions and functional forms without prior model assumptions; validated on synthetic traffic models and published in *J. Phys.: Complexity* (peer-reviewed)
- Supervised an undergraduate team on hardware integration and software testing for experimental validation

Professional Experience

Visiting Assistant Professor, ERAU – Daytona Beach, FL Spring 2026

- Built and deployed a custom LLM-agent system integrating email monitoring, Canvas API retrieval, anonymized context tracking, and SMS-alerted draft replies, sustaining sub-2-hour response times across 140 students for a full semester
- Taught 4 sections across Dynamics, Uncrewed Systems, and Mechatronics; developed original materials for Dynamics and modernized inherited curricula for the remaining courses

Imaging Systems Specialist, ERAU – Daytona Beach, FL 2017 – 2020

- Led OCR-based document workflow automation, coordinating between internal stakeholders and an external development team
- Integrated admissions and registrar systems to streamline student document processing, reducing manual work from hours to minutes

Technology Project Manager, FDOT - Turnpike – Ocoee, FL 2014 – 2017

- Managed development of a GIS-integrated road-inspection dashboard unifying multiple legacy systems for statewide per-mile visual inspection, photo history, and maintenance tracking
- Coordinated bi-weekly development cycles of a third-party development firm

Senior Technical Roles 2009 – 2014

Apple Inc., Stetson University

- Managed enterprise IT infrastructure, technical troubleshooting, and multimedia systems across multiple organizations, building foundational expertise in systems integration, documentation, and customer support.

Publications & Presentations

- D. Lane, "A Study in Object Detection and Classification Performance by Sensing Modality for Autonomous Surface Vessels" (2025). Doctoral Dissertations and Master's Theses. 937. https://commons.erau.edu/edt/937.
- D. Lane and S. Roy, "Validating a data-driven framework for vehicular traffic modeling," *Journal of Physics: Complexity*, vol. 5, no. 2, p. 025008, May 2024. doi: 10.1088/2632-072X/ad3ed6.
- D. Lane and S. Roy, "Using information theory to detect model structure with application in vehicular traffic systems," *IFAC-PapersOnLine*, vol. 56, no. 3, pp. 367–372, 2023. 3rd Modeling, Estimation and Control Conference (MECC 2023). doi: 10.1016/j.ifacol.2023.12.051.
- D. Lane and S. Roy, "Creating Data-Driven Model Discovery Framework with Applications in Traffic Modeling," poster presentation, SIAM Conference on Applications of Dynamical Systems (DS23), Portland, OR, May 2023.
- D. Lane and E. Coyle, "Appendix D: Vision System," in *Minion: Design and Competition Strategy for the 2024 Maritime RobotX Challenge*, Team Minion Technical Design Report, RoboNation, 2024. Available: robonation.org [p. 30]
- D. Lane, "Aligning Big Data Management with Business Priorities," Florida Transportation Data Symposium, 2015. [slides]

Selected Projects

Graduate projects across ML, multi-agent systems, and motion planning. Documentation at www.dplane.com.

- **PongBot — Autonomous Targeting Robot:** Autonomous robot using computer-vision target detection and a stepper-driven pan/tilt launcher; reliably lands projectiles in targets at 6 m
- **Neural-Network System Identification:** Identified nonlinear systems (harmonic, Van der Pol oscillators) with neural networks; evaluated hyperparameter sensitivity and sparsity-penalized / genetic-algorithm alternatives
- **Multi-Agent Consensus Dynamics:** Compared binary and multivariate voter models for network consensus, quantifying how graph topology and parameters govern convergence rate and stability
- **Motion Planning Under Kinodynamic Constraints:** Adapted grid-based A^* and sampling-based RRT^* for a differentially-steered robot with jumping capability; jumping improved path efficiency in dense obstacle fields, with A^* yielding optimal paths and RRT^* faster computation